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Title: ***The impact of green and blue space on general health in local authorities in England: Sex differences revealed through multiple regressions***

2. Abstract

This project provides a comprehensive assessment of the influence of the amount of green and blue space per 1000 people in a local authority in England on general health outcomes, highlighting differences by sex. Using data that is publicly available from the Rivers Trust and the Office for National Statistics (ONS), it was found that there is a positive association between green and blue space and general health with differences in this effect for men and women.

3. Introduction

RQ1: Does the percentage of green and blue space in a local authority predict general health?

H0: There is no effect of percentage of green and blue space in a local authority on general health

H1: There is a positive effect of the percentage of green and blue space in a local authority on general health

RQ2: Does the effect (if any) or green and blue space on general health differ by sex?

H0: The effect of percentage of green and blue space on general health does not differ by sex

H1: The effect of percentage of green and blue space on general health differs by sex

RQ3: Does sex affect the relationship between percentage of green and blue space in a local authority and general health?

H0: There is no effect of sex on the relationship between percentage of green and blue space in a local authority and general health

H1: There will be an interaction effect of sex on the relationship between percentage of green and blue space in a borough and general health

Nature has been widely shown to have a positive impact on health outcomes (Jimenez et al., 2021). Research shows that spending time in green spaces reduces risk of chronic disease such as cardiovascular disease and diabetes as well as improving mental health, while blue space lowers obesity rates and improves emotional wellbeing. (Wang et al., 2025). For our project we were particularly interested in looking at how green and blue space availability at a local authority level influences general health outcomes in the same area and whether these effects differ by sex.

In this paper we look at how the amount of green and blue space per 1000 people in a local authority in England is correlated with general health outcomes in the same area. We construct a model in which we find that general health outcomes are partially explained by the amount of green and blue space, and that males, on average, score higher on general health measures. Future works could look at the effect of green and

blue space separately on general health outcomes, which would allow a more nuanced understanding of these effects.

The remainder of this paper is structured as follows: Section 4 describes the data sources and variables used and section 5 outlines the regression models and analysis plan. Section 6 presents the key results including robustness checks. Finally, Section 7 discusses our findings, limitations of the project and next steps.

4. Data

For our project we will use two publicly available datasets to investigate the relationship between green and blue space and general health outcomes in local authorities in England.

The first dataset is from the **Rivers Trust** (*Area of Accessible Green and Blue Space per 1000 Population (England)*, n.d.) and reports the area of publicly accessible green and blue space by local authority. Green space is made up of public parks, nature reserves etc., whereas blue space is an open area with water such as a rivers or lake. The data is normalised per 1000 people in order to control for differences in populations between local authorities.

The second dataset is from the **2021 Census** obtained the **Office for National Statistics** (*General Health by Age, Sex and Deprivation, England and Wales - Office for National Statistics*, n.d.). This reports the self-reported health status for people in each local authority split into five categories of very good, good, fair, bad, and very bad. In order to be able to use this data for our analysis we calculated a weighted average health score for each local authority. This weighted health score takes into account the percentage of people who report each different health category with higher health scores signifying better general health.

Within the 2021 Census dataset the results for health status are also split by sex, so we can compare the potential differences between men and women. Datasets for the Index of Multiple Deprivation (IMD) (*English Indices of Deprivation 2019*, n.d.), urban or rural classification for each local authority (*Local Authority Districts (December 2021) Boundaries EW BFE and Rural Urban Classification*, n.d.) and life expectancy (proportion of life spent in good health) (*Healthy Life Expectancy, UK - Office for National Statistics*, n.d.) will also be used in robustness checks.

Data wrangling steps in Excel before uploading to code book

Before python we wrangled the **healthy life Expectancy dataset** (*Healthy Life Expectancy, UK - Office for National Statistics*, n.d.) in excel so that we could upload it. The following steps were completed:

1. Deleted all but sheet 1
2. On sheet 1, deleted all rows but column headers and Local Authority level, 2022-24 rows
    a. Deleted rows 101,968 and below
    b. Deleted rows 8 - 95,927
    c. Deleted rows 1 - 6

We also wrangled the **2021 Census general health dataset** (*General Health by Age, Sex and Deprivation, England and Wales - Office for National Statistics*, n.d.). The following steps were completed for this dataset:

1. Used only Table 5 tab (and deleted all other tabs)
2. Deleted rows 4941 onwards (2011 Data)
3. Deleted rows 1-4

5. Model

Following our initial scoping of the data, we realised that there were large differences in green and blue space per 1000, with some areas being as far as 10+ standard deviations away from the mean. Therefore, we decided to conduct a log-transformation of the green space data to cluster the data together.

Following this, we decided to run 3 regressions:

RQ1: A Simple Linear Regression of weighted health score predicted by the log-transformed green and blue space

RQ2: A Multiple Linear Regression of weighted health score predicted by log-transformed green and blue space + sex

RQ3: A Multiple Linear regression of weighted health score predicted by log-transformed green and blue space + sex + green and blue space interacted with sex

We chose sex as a variable as we believed it might provide an interesting analysis, particularly in the interaction between green and blue space and sex, where a difference would suggest green and blue space has a different effect on health for men compared to women. We also found research to highlight that women tend to report "significantly poorer health than men on all self-reported health indicators" (Boerma et al., 2016).

We also decided to run multiple robustness checks. Our first check involved a multiple linear regression including the addition of the IMD (Index of Multiple Deprivation) score and whether the Local Authority was classified as Urban or Rural. The second robustness check involved running the initial linear regression with life-expectancy data instead of weighted health score, as we wanted to see whether the effect (if any) was replicated with objective measures of health as well as subjective. We also ran a post-hoc power analysis to see how high the power of our study would've been in retrospect if it were to be conducted using primary data.

6. Results

RQ1:

Our estimates based on our minimal model suggest that a 1 km^2 increase in the log-transformed green space per 1000 people resulted in a 0.0204 increase in health score, $t_{602} = 9.482$, $p < 0.0001$. A post-hoc power analysis was run with the coefficients from the regression, and the power was outputted as 1.00, suggesting that retrospectively, our sample size was large enough for the given effect size to guarantee that the result we have found is not a false positive.

RQ2:

Based on the multiple regression model, being male leads to a 0.0203 increase in health score, a score which was independent of the log-transformed green and blue space per

1000 in a local authority, t_{601} = 3.30, p =0.001.

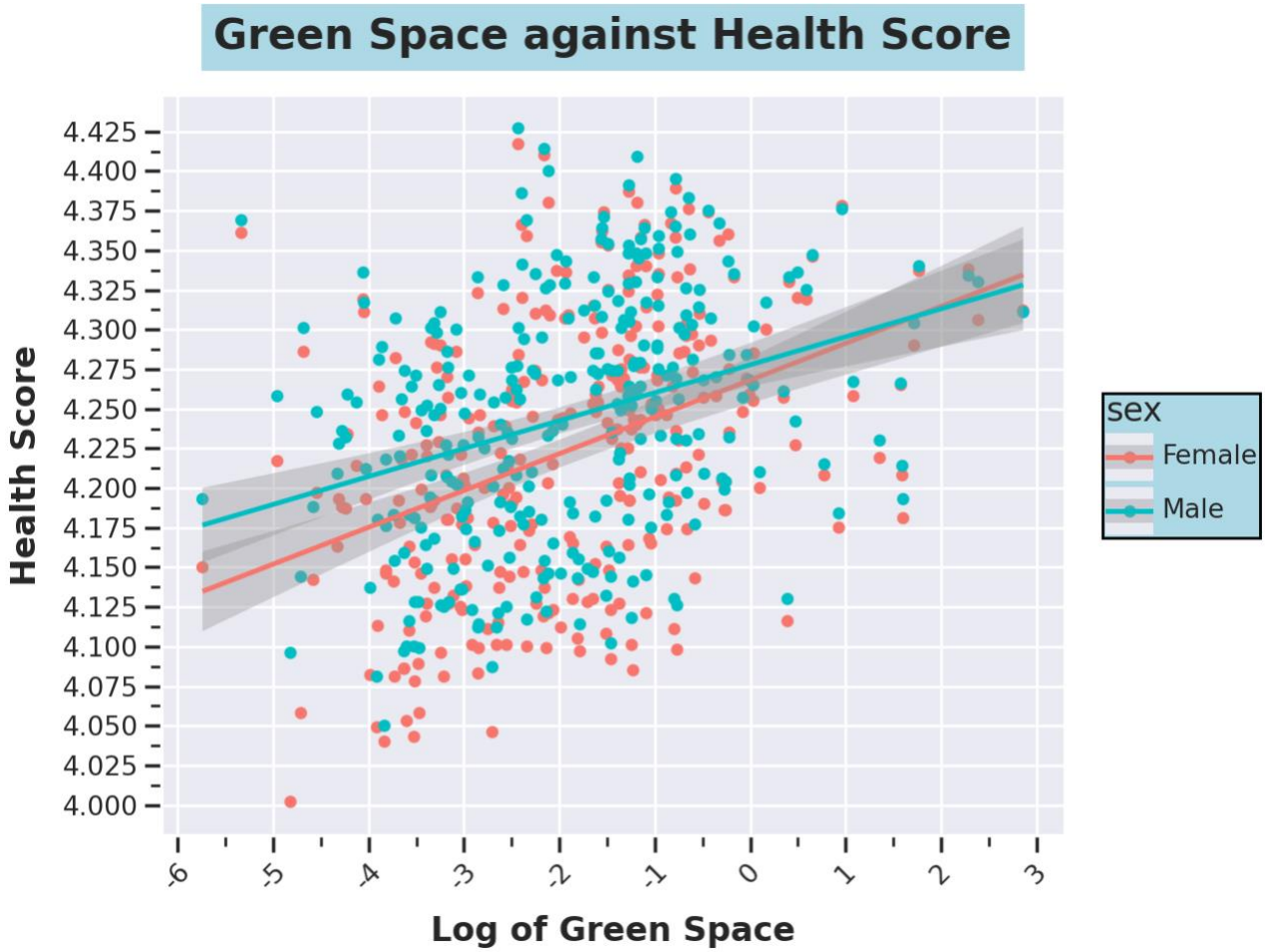

Figure 1.

Log-transformed green space against Health Score, where red data points are health score for each Local Authority for females, and blue data points are health score for each Local Authority for males.

RQ3:

Based on the third model, we did not find an interaction effect between sex and the log-transformed green and blue space per 1000, t_{600} = -1.331, p = 0.184.

Robustness Check 1:

For our first robustness check, we ran a multiple linear regression model, which included the controls of whether the local authority was classified as urban or rural, and the IMD (Index of Multiple Deprivation) score for the Local Authority. We found that the addition of these controls led to no statistically significant effect of the log-transformed green and blue space per 1000 on health scores, t_{599}= 1.473, p =0.141. We also found that whether

the town was rural or urban did not have an impact on health scores independent of the other variables, $t_{599} = 0.948$, $p = 0.344$.

Robustness Check 2:

For our second robustness check, we ran a simple linear regression model, using the data for life expectancy instead of self-reported health scores, and we found that a 1km^2 increase in log-transformed green space resulted in 0.5% increase in healthy life expectancy, $t_{230} = 1.973$, $p = 0.050$.

We also ran a Variance Inflation Factor (VIF) test, to check for collinearity in our model, the results of which are shown in the table below.

	variable	VIF
0	const	8.84
1	urban_int	1.69
2	log_gs	1.76
3	gender_int	1.00
4	imd_score	1.19

As all the variables have a VIF are close to 1, therefore we can suggest that the multiple linear regression model is suitable and there is minimal collinearity.

7. Discussion

Summary of findings

Our findings suggest that there is a positive relationship between green and blue space and general health scores, such that the more green and blue space within a Local Authority, the greater the self-reported health score. We also found a statistically significant difference in scores between the sexes, with men scoring higher than women. However, when accounting for the IMD score, the effect of green and blue space became statistically insignificant. This may be because 'Living Environment Deprivation' is used when calculating the IMD score (GOV.UK, 2025), which measures the quality of the local indoor and outdoor environment.

Limitations

There were several limitations to our project. Firstly, the data we used as a measure of general health outcomes was self-reported, meaning it may have been influenced by personal bias. For example, what may be seen as 'very good' for one person could be 'very bad' for another, based on previous circumstances or expectations.

Secondly, the data we used was for local authorities as a whole and therefore does not show variation within these areas. Ultimately using data from individual people would give a more accurate picture of how green and blue space affects health outcomes

Lastly, "access to beaches and the sea could not be factored into the analysis" (*Area of Accessible Green and Blue Space per 1000 Population (England)*, n.d.) for the green and blue space dataset. This would have skewed the data for blue space, especially for areas close to the sea. In order to better understand the relationship next time, we could use data that includes beaches as blue spaces in its recordings.

Ideas for future research

The data for green and blue space in our dataset was combined, next time we could use a dataset that splits these in order to access the relationship of green space on health and blue space on health independently.

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AI Declaration: LLMs used for initial scoping of datasets only. All other authorship was completed unassisted by group members.